NeuroSynIQ

AI-Assisted Neuromotor Monitoring System

ENGINEERING EVIDENCE PORTFOLIO | INNOVATION CATEGORY | 2026

Theme: AI — Engineering a Safer Future

01 Project purpose and safeguard

Problem: Changes in movement performance between medical appointments may be difficult to notice or describe consistently. NeuroSynIQ is a research prototype intended to record simple neuromotor measurements over time and highlight differences for a neurologist to review.

Human safeguard: The system explains what changed and checks whether changes persist. It does not diagnose neurological disease, determine a medical cause, prescribe treatment, or replace a doctor.

Engineering goal and measurable design criteria

Requirement	Prototype implementation / constraint
Accessible input	Tablet or computer; no additional specialized sensor required
Three repeatable tasks	Tap: 10 seconds; Reaction: 5 attempts; Trace: follow a fixed path
Personal reference	MotorPrint built from 3 Tap, 3 Reaction and 3 Trace baseline sessions
Consistency over time	Lock the established MotorPrint before new monitoring checks
Avoid premature persistence labels	PersistenceGuard: 1 observed; 2 repeated; 3+ persistent changes
Understandable results	WHY? panel shows baseline, current value and explanation
Storage	Firebase Firestore used for session history; access rules and offline behaviour require verification
Budget	TT$0 additional software-development cost using existing equipment and free tools; display and existing hardware costs excluded

What is innovative?

The innovation is the combination of existing movement tasks, an individual baseline, repeated-change decision logic, and plain-language explanations in one functioning research prototype—not the invention of tapping or reaction tests.

Student contributions and responsibilities

Eliana Jagesar — Software Development & Engineering
Developed and improved the NeuroSynIQ software prototype, including the movement tests, MotorPrint personal baseline, PersistenceGuard decision logic, and WHY? explanations, using AI-assisted development tools where applicable.

Elisa Maharaj — Research, Testing & Presentation
Contributed to project research, testing and evaluation, preparation of the science fair display, and presentation.

The project is presented collaboratively, with software development credited separately from research, testing, display preparation and presentation.

02 Development and iteration log

The milestones below summarize documented project work. The app screenshots in this portfolio were captured on 14 October 2026; they are evidence of the later working interface, not claimed to be screenshots from earlier dates.

Period	Development stage	Change / engineering reason
30 Sep 2026	Initial functional tests	Built Tap, Reaction and Trace tasks to capture measurable movement and response signals.
3–4 Oct 2026	Improved Trace measurement	Moved beyond a 100% corridor score to mean deviation, variability, completion time and largest deviation.
4–7 Oct 2026	Personal MotorPrint + lock	Used repeated sessions to calculate personal averages and variation; locking prevents new checks from shifting the reference.
4–7 Oct 2026	PersistenceGuard	Added consecutive-session logic to distinguish one-off, repeated and persistent measured changes.
4–7 Oct 2026	WHY? and Research Lab	Added understandable explanations and controlled simulated scenarios to evaluate the decision rule.

Current working prototype — genuine application screenshots

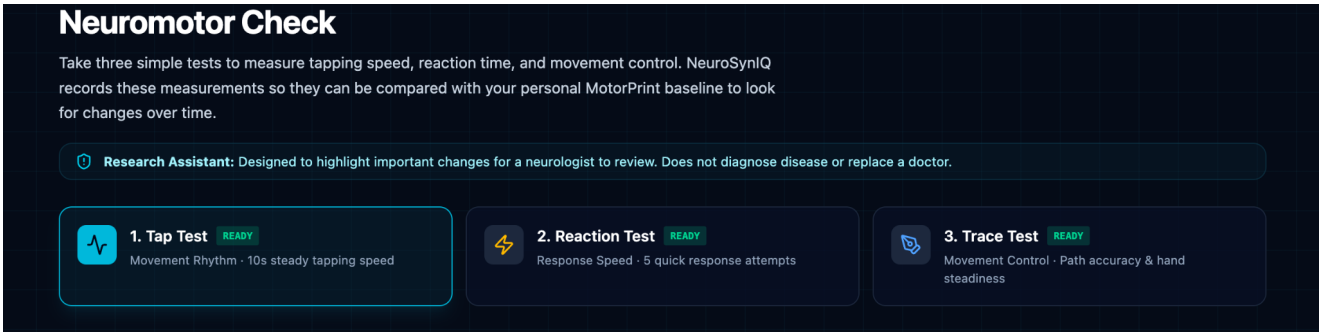


Figure 1. Three available tests in the actual NeuroSynIQ interface (screenshot captured 14 Oct 2026).

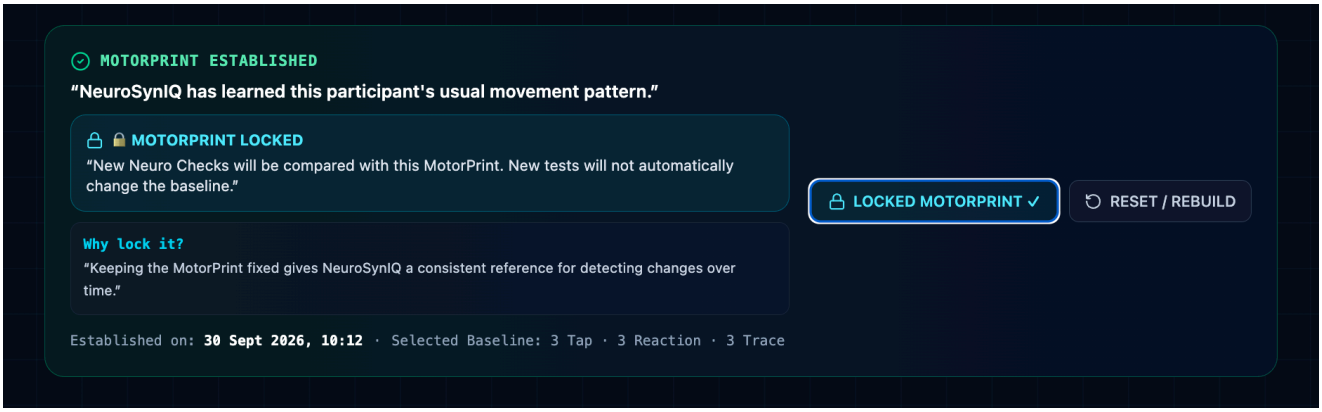


Figure 2. MotorPrint established and LOCKED; 3 sessions of each test selected. This is the fixed reference for subsequent checks.

03 Actual code excerpts and explanations

A. PersistenceGuard — count consecutive changes

```
// Count consecutive MEASURED_CHANGE sessions from the latest session backwards
let consecutiveChangeCount = 0;
for (let i = classifications.length - 1; i >= 0; i--) {
  if (classifications[i] === 'MEASURED_CHANGE') {
    consecutiveChangeCount++;
  } else {
    break;
  }
}
```

Student explanation: “The loop starts with the newest session. Every consecutive measured change increases the count by one. As soon as a session is not a measured change, the loop stops. The broader algorithm then uses the count to describe observed, repeated or persistent change.”

B. MotorPrint — calculate tap baseline and variation threshold

```
const tapBaselineVal = Math.round(motorPrint.tap.typicalTapCount.mean);
const tapStdDev = motorPrint.tap.typicalTapCount.stdDev || 2;
const threshold = Math.max(1, Math.round(tapStdDev));
```

Student explanation: “The first line rounds the personal average tap count. The next reads the standard deviation, with a fallback value of 2 if it is zero or unavailable. The final line makes the threshold at least 1. This excerpt calculates a threshold; the full code performs the comparison.”

Code provenance and student ownership

These excerpts are reproduced exactly from the code supplied for the project. The full GitHub repository is private. The student should be prepared to locate the corresponding files in the actual project and explain how AI-assisted coding tools were used, what they personally directed, tested and improved.

Engineering decision rule

Latest consecutive changed sessions	Interpretation
0	Within baseline / sequence interrupted
1	Change observed — persistence not established
2	Repeated change — monitoring recommended
3 or more	Persistent measured change — human review recommended

04 MotorPrint and WHY? — evidence from the app

MotorPrint is based on three sessions for each of the three tests. The established reference can be locked so new checks do not automatically change the baseline.

Measure	Personal baseline / variation
Tap count	36.7 taps; ±19.6 taps (usual range 17–56)
Reaction time	542 ms; ±123.4 ms (usual range 419–665)
Mean trace deviation	5 px; ±3.3 px (usual range 1.7–8.3)
Trace variability	±3 px; ±1.1 px
Trace completion time	14 s; ±4.7 s

WHY? result — current value compared with the baseline

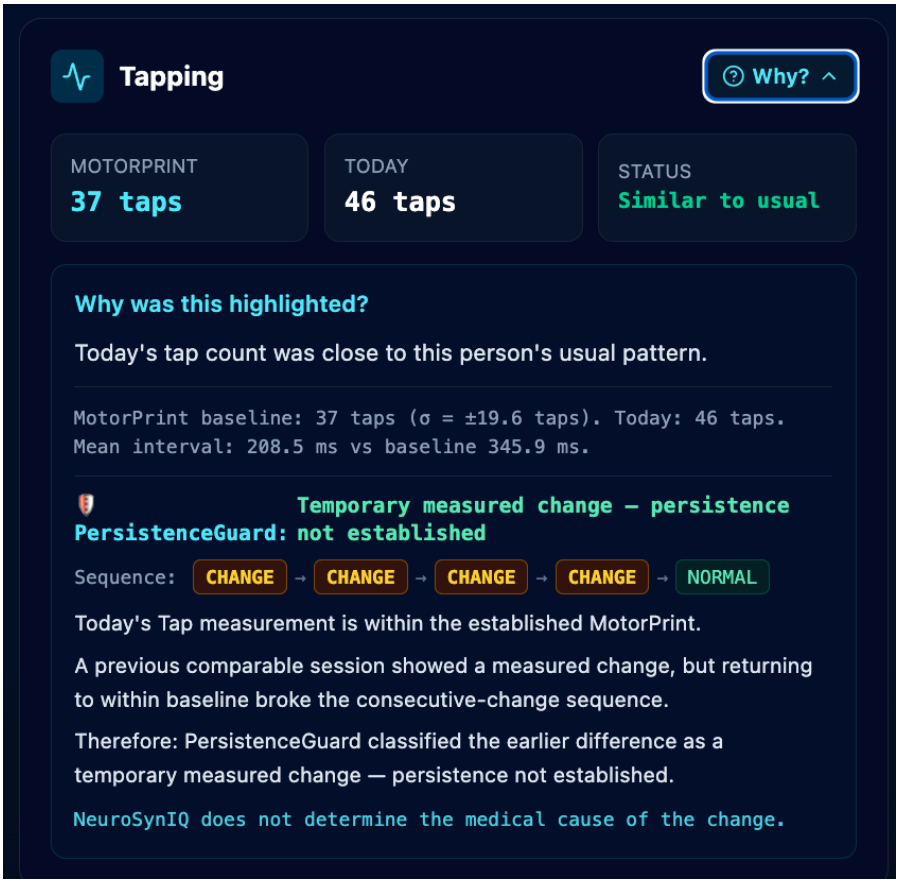


Figure 3. Genuine WHY? screen: 46 taps today versus 37 taps in MotorPrint. The result is similar to usual; the latest normal session interrupts an earlier changed-session sequence.

What should be noticed

A result can be within the person's baseline even when it differs numerically from the mean. The explanation panel gives the measurements, the interpretation and the repeated-change context rather than simply showing a warning icon.

Clinical boundary

A variation band is a prototype engineering reference, not a medically validated diagnostic threshold. Reaction time and tracing may be affected by fatigue, practice, device response, attention and test conditions.

05 Research Lab — 100 simulated cases

COMPUTER-GENERATED ALGORITHM SIMULATION — NOT CLINICAL VALIDATION. The 100 cases were simulated under five predefined scenarios, 20 cases per scenario, with five sessions each (500 simulated sessions total). They test the PersistenceGuard decision rule, not neurological diagnostic accuracy.

Scenario	Cases	Strict persistence confirmed within 5 sessions
Stable pattern	20	0
Normal variation	20	0
One-off temporary change	20	0
Gradual change	20	17
Persistent change	20	20
TOTAL	100	37

What was measured?

Simulation outcome	Observed result
Persistent / gradual cases with strict 3-session confirmation	37 of 40 = 92.5%
Persistent / gradual cases reaching repeated-or-persistent status	40 of 40 = 100%
One-off cases incorrectly called strictly persistent	0 of 20
Gradual cases not strictly confirmed within five sessions	3 of 20 gradual cases
Confirmation trade-off	Three consecutive changed sessions require additional observations

Interpretation and limitation

PersistenceGuard avoided strict persistence labels for all 20 one-off simulated cases. However, 3 gradual-change cases did not reach strict confirmation within the five-session window. A later session could be needed. These results show the behaviour of a rule under controlled inputs; they do not establish sensitivity, specificity or safety for real patients.

Additional empirical evidence to collect

For a stronger real-world test, record repeated actual Tap, Reaction and Trace measurements under consistent conditions, including device, timing, participant consent where appropriate, and sources of measurement variation. Keep real measurements clearly separate from simulation results.

06 Engineering trade-offs and safeguards

Decision	Benefit	Limitation / trade-off
Three tests	Captures different aspects of performance	Requires more participant time
Personal baseline from 3 sessions	Accounts for individual differences	Small baseline may be variable
Lock MotorPrint	Provides stable comparison reference	Needs a deliberate reset/rebuild when justified
PersistenceGuard 3-change rule	Avoids premature persistent labels for one-off changes	May delay confirmation of gradual change
WHY? explanations	Makes results understandable for review	Explanations do not identify a medical cause
Firebase Firestore	Stores session history for comparison	Security rules and offline behaviour require testing

Evidence note: This portfolio contains screenshots captured 14 Oct 2026, supplied code excerpts and documented simulated results. Earlier iteration dates are development milestones; attach actual historical commits or earlier screenshots if available.